

Agents and Agentic AI Assignments

Date given: Feb 06 2026

Assignment 1: Agents and Agentic AI

Get the Agents and Agentic AI Framework demos working, understand the code design

Download the below zip and place it in your genaitraining code folder:

https://github.com/vijay-agrawal/genai-batch3/blob/main/democode/agent_demos.zip

Get all the demos in this zip working in your environment and understand the functionality and code.

Assignment 2: Agentic AI Design Patterns

Download the below zip and place it in your genaitraining code folder:

https://github.com/vijay-agrawal/genai-batch3/blob/main/democode/agentic_designpatterns.zip

Explore the design patterns, run them. (Few of them are runnable as is – try to fix the others)

Assignment 3: Agentic RAG

Implement an Agent using ADK such that it can do ICD-10 lookup from supplied icd-10 to disease description JSON.

Use the following as your JSON data that provides ICD-10 code for disease based on description

https://github.com/vijay-agrawal/genai-batch3/blob/main/assignments/icd10_data.json

The tool should be able to answer the following questions:

Example Question 1

User Query: "Patient came in complaining their heart is racing and beating irregularly"

Why Semantic Search Helps:

- User describes symptoms ("heart racing," "beating irregularly") rather than using medical terms

- Semantic search understands these symptoms relate to cardiac rhythm problems
 - **Correct Match:** "arrhythmia" or "atrial fibrillation"
 - **ICD Code:** I49.9 (Cardiac arrhythmia, unspecified) or I48.91 (Atrial fibrillation)
 - Traditional keyword search for "heart racing" would fail to find these codes
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Example Question 2

User Query: "38-year-old woman with severe stomach acid coming up into throat, especially at night"

Why Semantic Search Helps:

- Describes symptoms ("stomach acid coming up into throat") in plain language
- Semantic search connects this symptom pattern to the underlying condition
- **Correct Match:** "acid reflux" or "gerd"
- **ICD Code:** K21.9 (Gastro-esophageal reflux disease without esophagitis)
- Keyword search wouldn't connect "acid coming up" with "reflux" or "GERD"

Example Question 3

User Query: "Child has itchy, watery eyes and keeps rubbing them - looks pink and crusty"

Why Semantic Search Helps:

- Uses descriptive symptoms ("pink," "crusty," "itchy," "watery eyes") rather than diagnosis
- Semantic search recognizes this symptom cluster as a common eye condition
- **Correct Match:** "pink eye" or "conjunctivitis"
- **ICD Code:** H10.9 (Unspecified conjunctivitis)
- Simple keyword matching wouldn't link "pink and crusty eyes" to conjunctivitis

Example Question 4

User Query: "Patient reports persistent sadness, loss of interest in activities, sleeping too much for past 6 weeks"

Why Semantic Search Helps:

Describes behavioral and emotional symptoms rather than using psychiatric terminology

Semantic search understands symptom duration and pattern matching mental health criteria

Correct Match: "depression" or "major depression"

ICD Code: F32.9 (Major depressive disorder, single episode, unspecified)

Traditional search wouldn't connect this symptom narrative to the specific diagnosis code

Implement an Agent (tool/function) that will retrieve them from a vector database so that benefits of semantic search can be realized.

Implementation Hints

- 1) Ensure your RAG demos are working correctly – they are the starting point – see:

https://github.com/vijay-agrawal/genai-batch3/blob/main/democode/rag_gemini_v2.zip

- 2) See agentic_designpatterns\demos\agentic_rag.zip

To see how retriever is used in Agent to achieve agentic RAG

Build your assignment by leveraging the above two. A streamlit app that demos the above 4 use-cases would be the best.

All the Best!